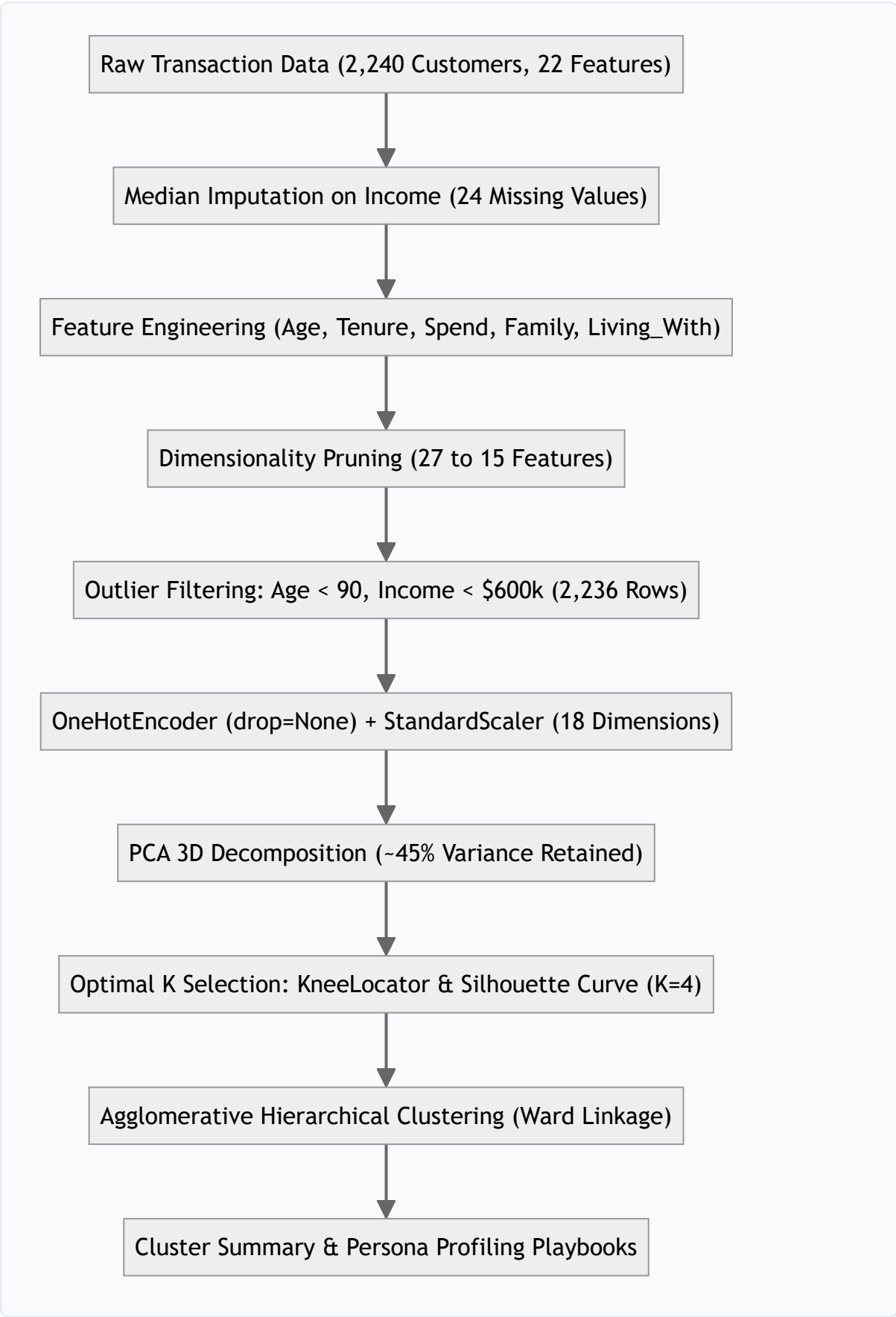
# SmartCart Customer Clustering System (Minor Project)

## Project Overview

The **SmartCart Clustering System** is an enterprise-grade unsupervised machine learning project designed to solve a critical commercial retail challenge: transitioning from indiscriminate mass marketing ("spray-and-pray") to personalized, data-driven customer segmentation.

By analyzing multi-dimensional customer transaction records, demographic structures, and purchasing channels, this system partitions an e-commerce customer base of 2,240 records into four high-cohesion, actionable behavioral personas.




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## 1. Project Directory & Assets

- **Jupyter Notebook**: `smartcart.ipynb` — 67 fully executable notebook cells detailing the complete pipeline.
- **Dataset**: `smartcart_customers.csv` — 2,240 raw customer records across 22 attributes.
- **Assignment Brief**: `SmartCart_Clustering_System.pdf` — Official minor project specifications and problem statement.

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## 2. Complete End-to-End Pipeline Architecture

### Phase 1: Exploratory Audit & Missing Data Imputation

- Dataset dimensions: 2,240 rows, 22 columns.
- `Income` missingness: Exactly 24 null values imputed using the distribution median ($M$) to withstand positive skewness and high-earner outliers.

### Phase 2: Feature Engineering & Domain Transformations

1. **`Age`**: Derived as $2026 - \text{Year_Birth}$.
2. **`Customer_Tenure_Days`**: Anchored against $\max(\text{Dt_Customer})$ rather than dynamic clock time to ensure pipeline reproducibility across runtime environments.
3. **`Total_Spending`**: Consolidated sum across 6 expenditure categories (`MntWines`, `MntFruits`, `MntMeatProducts`, `MntFishProducts`, `MntSweetProducts`, `MntGoldProds`).
4. **`Total_Children`**: Sum of young children (`Kidhome`) and teenagers (`Teenhome`).
5. **`Education` Mapping**: Simplified into 3 tiers (`Undergraduate`, `Graduate`, `Postgraduate`).
6. **`Living_With` Mapping**: Consolidated into `Partner` vs `Alone`.

### Phase 3: Dimensionality Pruning & Outlier Elimination

- Dropped 12 redundant/replaced columns (`ID`, `Year_Birth`, `Marital_Status`, `Kidhome`, `Teenhome`, `Dt_Customer`, and the 6 individual spending columns), reducing feature count to 15.
- Retained purchasing channels (`NumDealsPurchases`, `NumWebPurchases`, `NumCatalogPurchases`, `NumStorePurchases`).
- Filtered 4 extreme anomalies ($\text{Age} < 90$ and $\text{Income} < \$600{,}000$), preserving 2,236 clean customer records.

### Phase 4: Encoding & Standardization

- Encoded `Education` and `Living_With` via `OneHotEncoder(drop=None)` to preserve equidistant vector space geometry across categorical levels.

- Applied `StandardScaler` ($z = (x - \mu)/\sigma$) across all 18 feature coordinates.

### Phase 5: PCA & Hyperparameter Tuning

- Conducted 3-component PCA capturing **44.95%** of cumulative variance.
- Identified the convex elbow at $K = 4$ via `KneeLocator` on WCSS (Inertia).
- Corroborated $K = 4$ through silhouette score evaluation.

### Phase 6: Model Selection & Customer Profiling

- Evaluated K-Means vs Agglomerative Hierarchical Clustering.
- Selected **Agglomerative Clustering with Ward's linkage** due to superior geometric separation in 3D PCA projection.

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## 3. Production Cluster Summary Table ($N = 2,236$)

| Feature | Cluster 0 (Red) | Cluster 1 (Blue) | Cluster 2 (Yellow) | Cluster 3 (Green) | | :--- | :---: | :---: | :---: | :---: | | **Customer Population** | **905 (40.5%)** | **534 (23.9%)** | **444 (19.9%)** | **353 (15.8%)** | | **Household Income ($)** | **\$39,681** | **\$72,808** | **\$36,960** | **$70,723** | | **Total Spending ($)** | **\$222** | **\$1,237** | **\$166** | **$1,190** | | **Total Children** | **1.24** | **0.51** | **1.27** | **0.46** | | **Living With Partner** | **100%** | **100%** | 0.7% | **0.0%** | | **Living Alone** | **0.0%** | **0.0%** | **99.3%** | **100%** | | **NumStorePurchases** | 4.14 | **8.66** | 3.62 | **8.43** | | **NumCatalogPurchases** | 0.97 | **5.50** | 0.84 | **5.01** | | **NumWebVisitsMonth** | **6.31** | 3.58 | **6.66** | 3.73 | | **Deals Purchases** | **2.59** | 1.96 | **2.59** | 1.86 | | **Campaign Response Rate** | 7.6% | 16.7% | 14.2% | **32.0%** |

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## 4. Actionable Customer Persona Playbooks

1. **Cluster 0 — "Family Shoppers / Thrifty Couples"**: Target with bulk grocery value packs, weekend discount coupons, and family bundle promotions.
2. **Cluster 1 — "Affluent Couples / Established Spenders"**: Target with VIP loyalty programs, premium lifestyle catalogs, and early access to luxury collections.
3. **Cluster 2 — "Budget Singles / Window Shoppers"**: Target with clearance flash sales, cart abandonment vouchers, and zero-interest installment options.
4. **Cluster 3 — "Affluent Singles / Golden Cluster"**: Target with bespoke 1-on-1 concierge recommendations, gourmet subscriptions, and high-margin impulse offers (Highest ROI).